

LRFShap: Low-Rank Fine-Tuning-Free Data Shapley for Scalable Attribution

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Abstract

As the volume of data grows rapidly, data valuation has become an important tool for assessing the value of individual data points. It has been used in various practical scenarios, including data pricing, dataset curation, and selecting informative samples for training large-scale models such as large language models (LLMs). A major line of work in this direction is based on the Shapley value, which provides a principled framework for data valuation. However, its high computational cost, caused by repeated evaluation over many training subsets, limits its applicability to large-scale settings. FreeShap (Wang et al., 2024a) addresses this bottleneck by using the empirical Neural Tangent Kernel (eNTK) to replace repeated model fine-tuning with kernel ridge regression. In this work, we propose *Low-Rank Fine-tuning-free Shapley value (LRFShap)*, which further accelerates FreeShap by exploiting the eigenspectral structure of the eNTK matrix. LRFShap uses a low-rank eigendecomposition-based approximation of the eNTK matrix, achieving up to a 20 $\times$ speedup over FreeShap while preserving comparable selection accuracy when selecting a small fraction of valuable samples. We also provide theoretical support for our method and a spectral quantitative analysis explaining why the effect of rank on data selection varies across datasets.

1. Introduction

Data valuation has been studied for a long time. In practice, data valuation is used for pricing and trading datasets in data marketplaces, allocating rewards to data contributors, and curating training data by identifying valuable, redundant, or harmful examples (Agarwal et al., 2019; Dao et al., 2018; Hynes et al., 2018; Andrés Azcoitia & Laoutaris, 2022).

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Moreover, with the advent of large language models (LLMs) (OpenAI, 2023; Huang et al., 2024; Team et al., 2023), data valuation has become an increasingly important problem. It is well known that the performance of pre-trained foundation models is strongly affected by the quality of training data, much of which is collected through web crawling. For example, models in the Phi series (Gunasekar et al., 2023; Javaheripi et al., 2023; Abdin et al., 2024), trained on carefully curated data, demonstrate strong coding ability despite their relatively small size compared with other foundation models. This motivates scalable data valuation methods that can identify useful training samples from large datasets.

A common approach to data valuation in machine learning is based on the Shapley value (Shapley, 1953), which measures the contribution of each player in a cooperative game. Among Shapley value-based approaches, *Data Shapley* (Ghorbani & Zou, 2019; Jia et al., 2019) treats each training sample as a player and computes its contribution. However, many Data Shapley methods are difficult to use in practical machine learning settings because they require an exponential number of model retraining steps. Existing estimators reduce the combinatorial cost of Shapley value estimation using Monte Carlo sampling, group testing, randomized experiments, or variance reduction (Ghorbani & Zou, 2019; Jia et al., 2019; Lin et al., 2022; Wu et al., 2023). However, these methods still require evaluating many coalition utilities, which typically involves retraining or fine-tuning a model on each sampled subset. Data Utility Learning reduces this retraining cost by learning a surrogate utility predictor from a limited number of retrained coalitions (Wang et al., 2021), while FreeShap (Wang et al., 2024a) replaces subset fine-tuning with eNTK-based kernel regression. Although FreeShap avoids repeated model retraining by precomputing the eNTK matrix, each prefix in the Truncated Monte Carlo (TMC) still requires solving a subset kernel ridge regression. Our work targets this remaining kernel-solve bottleneck by replacing the full subset kernel system with a low-rank feature-space solve.

In this work, we propose *Low-Rank Fine-tuning-free Shapley value (LRFShap)*, which further accelerates eNTK-based Shapley value evaluation via low-rank kernel approximation. Our contributions are as follows: (i) we show that LRFShap substantially speeds up Shapley value evaluation while preserving the ability to select valuable samples in the

small-data selection regime; and (ii) we provide theoretical support for our method and a spectral quantitative analysis explaining why the effect of rank on data selection varies across datasets.

2. Related work

Data Shapley and its approximations. Data Shapley (Ghorbani & Zou, 2019; Jia et al., 2019) formulates data valuation as a cooperative game by treating each training sample as a player. The Shapley value (Shapley, 1953) assigns each data point a value according to a predefined utility, such as test loss or accuracy. Although this gives a principled framework for data valuation, its combinatorial computational cost leads to prohibitive computational cost, limiting its use in practical machine learning settings. To reduce this cost, prior work has proposed Monte Carlo, permutation-based, group-testing, and variance-reduced estimators for Shapley values (Mitchell et al., 2022; Burgess & Chapman, 2021; Wu et al., 2023; Illés & Kerényi, 2019). These estimators improve scalability, but Shapley value estimation can still be too expensive for data selection in large-scale datasets.

Reducing retraining costs in data Shapley. To reduce the retraining cost in Data Shapley, a line of work proposes computing Shapley values by avoiding or reusing model retraining. FreeShap (Wang et al., 2024a), the main baseline of this paper, uses eNTK-based kernel regression to avoid subset fine-tuning when approximating Shapley values. Wang et al. (2024b) estimate data values through a single training run using gradient and Hessian information. Sun et al. (2026) train a reusable explainer to estimate the value of each training sample. Data Utility Learning (Wang et al., 2021) retrains models on a limited number of sampled coalitions and learns a utility predictor to evaluate the remaining samples.

Low-rank kernel approximation for scalable kernel regression. Since LRFShap relies on a low-rank approximation of the eNTK, it is also related to low-rank kernel approximation for scalable kernel regression. Classical low-rank kernel approximation methods include eigendecomposition and the Nyström method (Golub & Van Loan, 2013; Williams & Seeger, 2000). Cortes et al. (2010) analyze how kernel approximation error affects learning performance, and their analysis is used in our theoretical understanding. Bach (2013) use the effective dimension to study when an approximate kernel method can perform similarly to the full kernel method.

3. Preliminaries

We introduce the notation for Data Shapley (Ghorbani & Zou, 2019; Jia et al., 2019), which serves as the data valuation framework used throughout this paper. We then review the computational challenges of Data Shapley and the eNTK-based FreeShap (Wang et al., 2024a) that motivates our low-rank approach.

3.1. Setup and Notation

Let $D_N := \{(x_i, y_i)\}_{i \in N}$ denote the training dataset indexed by $N := \{1, \dots, n\}$, where $x_i \in \mathbb{R}^d$ and the class label y_i is represented as a one-hot vector in $\{0, 1\}^C$. Similarly, let $D_T := \{(x_i, y_i)\}_{i \in T}$, denote the test dataset, where T is the test index set.

For a subset $S \subseteq N$, let $D_S := \{(x_i, y_i)\}_{i \in S}$ denote the corresponding sub-dataset. For $S = \{i_1, \dots, i_{|S|}\} \subseteq N$, under any fixed ordering, define $X_S := [x_{i_1}, \dots, x_{i_{|S|}}]^\top \in \mathbb{R}^{|S| \times d}$, and $Y_S := [y_{i_1}, \dots, y_{i_{|S|}}]^\top \in \{0, 1\}^{|S| \times C}$. The ordering is immaterial and is used only to define the matrix representation. The test matrices $X_T \in \mathbb{R}^{|T| \times d}$ and $Y_T \in \{0, 1\}^{|T| \times C}$ are defined analogously.

3.2. Shapley Value

A transferable utility (TU) cooperative game is a pair (N, U) , where $N = \{1, \dots, n\}$ is the set of players and $U : 2^N \rightarrow \mathbb{R}$ is a utility function with $U(\emptyset) = 0$. For $S \subseteq N$, $U(S)$ denotes the utility generated by the players in S . The Shapley value distributes the total utility $U(N)$ among individual players by averaging their marginal contributions over coalitions that do not contain the player. For player $i \in N$, it is defined as

$$\phi_i(U) = \frac{1}{n} \sum_{l=0}^{n-1} \frac{1}{\binom{n-1}{l}} \sum_{\substack{S \subseteq N \setminus \{i\} \\ |S|=l}} (U(S \cup \{i\}) - U(S)).$$

Equivalently, the Shapley value is the expected marginal contribution of player i over a uniformly random permutation of the players:

$$\phi_i(U) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} (U(P_i^\pi \cup \{i\}) - U(P_i^\pi)), \quad (1)$$

where P_i^π denotes the set of players appear before i in the permutation π . The Shapley value is the unique allocation rule satisfying the axioms of efficiency, symmetry, dummy player, and additivity (Shapley, 1953). Therefore, it is widely regarded as a fair and principled method for distributing the total utility among players.

3.3. Data Shapley

We focus on the supervised learning setting. Given D_N and D_T , for any $S \subseteq N$, let $D_S \subseteq D_N$ be the corresponding training subset and f_S denote the model trained on D_S . *Data Shapley* (Ghorbani & Zou, 2019; Jia et al., 2019; Wu et al., 2023) treats each training sample as a player and defines a utility function $U(S)$ that evaluates the performance of f_S on D_T . Typical choices of utility function include the test accuracy $U_{\text{acc}}(S) := \frac{1}{|T|} \sum_{(x_j, y_j) \in D_T} \mathbf{1}[f_S(x_j) = y_j]$ and the negative test error $U_{\text{err}}(S) := -\frac{1}{|T|} \sum_{(x_j, y_j) \in D_T} (f_S(x_j) - y_j)^2$. The Shapley value assigns each data point a score that quantifies its contribution with U .

As shown in Equation 1, computing the Shapley value exactly requires averaging over all $n!$ permutations of training samples. Moreover, for each subset $S \subseteq N$, evaluating the marginal contribution $U(S \cup \{i\}) - U(S)$ typically requires training (or retraining) a model f_S and $f_{S \cup \{i\}}$ on S and $S \cup \{i\}$, respectively.

Thus, the Data Shapley framework has two main computational bottlenecks:

1. **Combinatorial complexity:** the intrinsic the intrinsic cost of averaging marginal contributions over all permutations.
2. **Model retraining cost:** the need to retrain a separate model for each subset S to evaluate its utility on S .

The first bottleneck is commonly mitigated by Monte Carlo sampling based on the permutation formulation in Equation 1, which gives an unbiased estimator of the Shapley value (Maleki et al., 2013). In this work, we focus on the second bottleneck, which remains particularly challenging for large neural networks.

3.4. Neural Tangent Kernel

The Neural Tangent Kernel (NTK) provides a theoretical framework for describing the training dynamics of infinitely wide neural networks through kernel regression (Jacot et al., 2018; Arora et al., 2019; Lee et al., 2019). For finite-wide networks, its empirical counterpart, the empirical NTK (eNTK), has also been used as a practical tool for analyzing neural networks in tasks such as pruning and active learning (Wang et al., 2023; Hemachandra et al., 2023). Prior work also shows that eNTK regression at pretrained weights can approximate prompt-based fine-tuning behavior in language and vision models (Malladi et al., 2023; Wei et al., 2022).

Let $f^1(x; \theta_0)$ denote one output logit of a pretrained model with parameters θ_0 . In the paper, for computational simplicity, we use a scalar single-logit eNTK: for any pair of inputs

$$x_1, x_2 \in \mathbb{R}^d,$$

$$\hat{\Theta}_{\theta_0}(x_1, x_2) := \left[\frac{\partial f^1(x_1; \theta_0)}{\partial \theta_0} \right] \left[\frac{\partial f^1(x_2; \theta_0)}{\partial \theta_0} \right]^\top. \quad (2)$$

For a coalition $S \subseteq N$, we define the eNTK ridge-kernel regression predictor trained on the corresponding training subset D_S and evaluated on the test set X_T as

$$f_S^{\text{eNTK}}(X_T) = k(X_T, X_S) (k(X_S, X_S) + \rho I_{|S|})^{-1} Y_S, \quad (3)$$

where $k(X_T, X_S)$ and $k(X_S, X_S)$ denote the eNTK test-train and train-train kernel matrices, respectively, defined as

$$k(X_T, X_S) = [\hat{\Theta}_{\theta_0}(x_t, x_j)]_{t \in T, j \in S},$$

$$k(X_S, X_S) = [\hat{\Theta}_{\theta_0}(x_i, x_j)]_{i, j \in S}.$$

We use this scalar single-logit kernel for computational efficiency. The full multi-output eNTK formulation and its scalar approximation are provided in Appendix D.2.

3.5. Fine-Tuning-Free Shapley Value

FreeShap (Wang et al., 2024a) effectively addresses the retraining bottleneck of Data Shapley by replacing subset fine-tuning with the eNTK ridge predictor in Equation 3. It follows the Truncated Monte Carlo (TMC) estimator (Ghorbani & Zou, 2019): for a sampled permutation π^t , FreeShap scans the prefixes $S_p^t = \{\pi^t[1], \dots, \pi^t[p]\}$, evaluates $u_p^t = U(S_p^t, D_T \mid f_{S_p^t} = f_{S_p^t}^{\text{eNTK}})$, and updates the Shapley estimate using the marginal gain $u_p^t - u_{p-1}^t$. Once the full eNTK matrices $K_{\text{tr}} = k(X_N, X_N)$ and $K_{\text{te}} = k(X_T, X_N)$ are precomputed, all coalition utilities can be evaluated by kernel operations without additional fine-tuning. The full pseudocode is provided in Appendix D.3.

The remaining bottleneck is solving the subset kernel systems. For each prefix S_p^t of size p , FreeShap applies Equation 3 with $S = S_p^t$, which requires solving a $p \times p$ kernel ridge regression system. In a basic implementation, solving these systems for all prefixes costs $\sum_{p=1}^n O(p^3) = O(n^4)$ per TMC permutation, or $O(Mn^4)$ over M permutations. Even with incremental inverse updates, the computation still operates on full prefix-size kernel systems. This remaining kernel-solve bottleneck motivates our low-rank feature-space approximation in Section 4.

4. Methodology

In this section, we propose *Low-Rank Fine-Tuning-Free Shapley value (LRFShap)*, which further accelerate the FreeShap by replacing full subset kernel solves with low-rank feature-space solves.

4.1. Low-Rank Fine-tuning-free Shapley value

Let $k(X_N, X_N) \in \mathbb{R}^{n \times n}$ denote the full training kernel matrix, and let $k(X_T, X_N)$ be the corresponding test-train kernel matrix formed from the eNTK matrix defined in equation 3. Assume that $k(X_N, X_N)$ admits the eigendecomposition $k(X_N, X_N) = U\Lambda U^\top$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. Let $U_r \Lambda_r U_r^\top$ be the best rank- r approximation of $k(X_N, X_N)$ formed by the top- r eigenpairs. Define the training feature matrix $\Phi_{\text{tr}} := U_r \Lambda_r^{1/2} \in \mathbb{R}^{n \times r}$. Similarly, define the test feature matrix $\Phi_{\text{te}} := k(X_T, X_N) U_r \Lambda_r^{-1/2} \in \mathbb{R}^{|T| \times r}$. For any subset $S \subseteq N$, let $\Phi_S \in \mathbb{R}^{|S| \times r}$ denote the submatrix of Φ_{tr} consisting of the rows indexed by S . Then $k(X_S, X_S) \approx \Phi_S \Phi_S^\top$ and $k(X_T, X_S) \approx \Phi_{\text{te}} \Phi_S^\top$. Using the Sherman–Morrison–Woodbury identity, we obtain the following low-rank approximation of the eNTK predictor in equation 3:

$$f_S^{\text{LRNTK}}(X_T) \approx \underbrace{\Phi_{\text{te}}^\top}_{|T| \times r} \underbrace{(\Phi_S^\top \Phi_S + \rho I)^{-1}}_{r \times r} \underbrace{\Phi_S^\top}_{r \times |S|} \underbrace{Y_S}_{|S| \times C}. \quad (4)$$

Algorithm 1 LRFSHap (Ours)

Require: Training set D_N , test set D_T , utility function U , maximum MC iterations M , eNTK K , regularization $\rho > 0$, eigen rank r .

Ensure: Shapley values $\phi_1, \dots, \phi_n$

- 1: Initialize $\phi_i \leftarrow 0$ for $i = 1, 2, \dots, n$.
 - 2: eNTK matrices $K_{\text{tr}} \leftarrow k(X_N, X_N)$, $K_{\text{te}} \leftarrow k(X_T, X_N)$
 - 3: Best rank- r approximation $K_{\text{tr}} \approx U_r \Lambda_r U_r^\top$
 - 4: Low-rank features $\Phi_{\text{tr}} \leftarrow U_r \Lambda_r^{1/2}$, $\Phi_{\text{te}} \leftarrow K_{\text{te}} U_r \Lambda_r^{-1/2}$
 - 5: **for** $t = 1$ to M **do**
 - 6: Sample a permutation π^t of N
 - 7: $u_0^t \leftarrow U(\emptyset, D_T)$
 - 8: **for** $p = 1$ to n **do**
 - 9: $S_p^t \leftarrow \{\pi^t[1], \dots, \pi^t[p]\}$
 - 10: Extract submatrix $\Phi_{S_p^t}$ from Φ_{tr} indexed by S_p^t
 - 11: Compute ridge predictor
 - 12: $f_{S_p^t}^{\text{LRNTK}}(X_T) \leftarrow \Phi_{S_p^t}^\top (\Phi_{S_p^t}^\top \Phi_{S_p^t} + \rho I_r)^{-1} \Phi_{S_p^t}^\top Y_{S_p^t}$
 - 13: Update Shapley estimate $\phi_{\pi^t[p]} \leftarrow \frac{t-1}{t} \phi_{\pi^t[p]} + \frac{1}{t} (u_p^t - u_{p-1}^t)$
 - 14: **end for**
 - 15: **end for**
-

As shown in line 11 of Algorithm 1, the ridge regression is solved in a reduced r -dimensional feature space. For a prefix subset S_p^t with $|S_p^t| = p$, computing the Gram matrix $\Phi_{S_p^t}^\top \Phi_{S_p^t}$ requires $\mathcal{O}(pr^2)$ operations, while inverting the resulting $r \times r$ matrix costs $\mathcal{O}(r^3)$. Hence, the computational cost per prefix is $\mathcal{O}(pr^2 + r^3)$.

Summing over all prefixes $p = 1, \dots, n$, the cost of a single Monte Carlo iteration is $\sum_{p=1}^n \mathcal{O}(pr^2 + r^3) = \mathcal{O}(n^2 r^2 + nr^3)$. Over M independent Monte Carlo permutations, the

overall computational complexity is $\mathcal{O}(M(n^2 r^2 + nr^3))$. When $r \ll n$, this is substantially smaller than the $\mathcal{O}(Mn^4)$ cost of the original FreeShap procedure. A detailed complexity analysis of the algorithms is provided in Appendix F.

4.2. Theoretical understanding

In this subsection, we provide theoretical support for the proposed method for which proofs are deferred to Appendix E. We assume the kernel regime (see Equations 13 and 14), under which training a neural network is equivalent to solving a kernel regression problem with a fixed eNTK. Accordingly, rather than analyzing parameter-space dynamics, we study the gradient flow induced by this fixed kernel.¹

Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a positive semi-definite kernel with associated reproducing kernel Hilbert space (RKHS) (Berlinet & Thomas-Agnan, 2011) $\mathcal{H}_k$, and assume that $k_{\max} := \sup_x k(x, x) < \infty$. For a subset S , let $f_{t,\rho}^S \in \mathcal{H}_k$ denote the gradient flow solution of Equation 20 at time t , initialized at $f_0 = 0$ with regularization parameter $\rho > 0$. Define

$$A_{S,\rho} := k(X_S, X_S) + \rho |S| I_{|S|}.$$

Following Appendix C of Haas et al. (2023), this gradient-flow solution admits the explicit representation

$$f_{t,\rho}^S(x) = k(x, X_S) \left(I_{|S|} - e^{-\frac{2t}{|S|} A_{S,\rho}} \right) A_{S,\rho}^{-1} Y_S. \quad (5)$$

This expression encompasses several classical kernel regression estimators; see Appendix E.1 for details.

We now analyze the validity of FreeShap (Algorithm 2) by studying whether the corresponding Shapley values remain stable along the gradient flow, i.e., whether they remain well-defined and consistent under gradient-based dynamics. Specifically, we show that the Shapley values induced by the kernel gradient flow converge exponentially fast to their infinite-time counterparts. This provides theoretical support for the use of FreeShap as an approximation to Data Shapley obtained from neural network training under the kernel regime.

Proposition 4.1. Assume $\rho > 0$. Let D_N be a training dataset and D_T be a test dataset. For each subset $S \subseteq N$, define the utility $U_{t,\rho}(S) := -\frac{1}{|T|} \|Y_T - f_{t,\rho}^S(X_T)\|_2^2$, where $f_{t,\rho}^S$ denotes the gradient flow solution at time t with regularization ρ given in Equation 5. Then, for each player $i \in N$, the corresponding Shapley values satisfy

$$|\phi_i(U_{t,\rho}) - \phi_i(U_{\infty,\rho})| \leq \frac{4k_{\max}}{\sqrt{\rho}} \left(1 + \frac{k_{\max}}{\sqrt{\rho}} \right) e^{-2t\rho}. \quad (6)$$

¹Although the eNTK is originally defined for the multiclass setting (see Equation 12), we restrict attention to a single output dimension, that is, assume $C = 1$, resulting in a scalar-valued kernel (see Equation 3 and Appendix D.2).

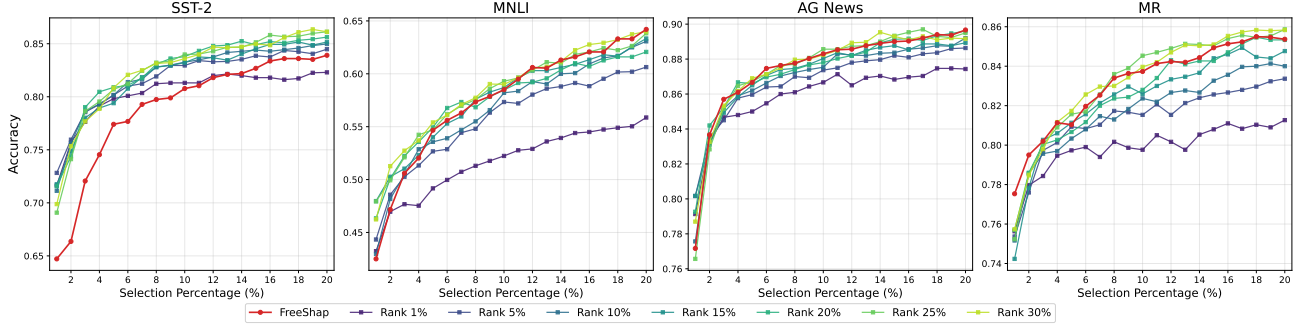


Figure 1. Prediction accuracy on four datasets across selection percentages. FreeShap is the baseline, and Rank $r\%$ denotes LRFShap using the top $r\%$ eigenpairs of the training eNTK matrix. Moderate ranks preserve accuracy and sometimes outperform the baseline.

The assumption $\rho > 0$ is crucial for obtaining the exponential convergence rate in Proposition 4.1. The ridgeless case $\rho = 0$ behaves differently, since the convergence rate depends on the smallest positive eigenvalues of the subset-dependent kernel matrices. We discuss this case separately in Appendix E.4.

Next, we study the approximation error introduced by LRFShap, which replaces the eNTK predictor Equation 3 with its low-rank predictor Equation 4 to improve computational efficiency. We quantify the resulting error in the corresponding Shapley values.

Proposition 4.2. Assume $\rho > 0$. Let D_N be a training dataset and D_T a test dataset. Let $\tilde{k}(X_N, X_N)$ be an approximation of $k(X_N, X_N)$, and assume that both matrices are symmetric positive semidefinite. For each subset $S \subseteq N$, define $\tilde{A}_{S,\rho} := \tilde{k}(X_S, X_S) + \rho|S|I_{|S|}$. Define

$$U_{\infty,\rho}(S) := \frac{1}{|T|} \|Y_T - f_{\infty,\rho}^S(X_T)\|_2^2,$$

$$\tilde{U}_{\infty,\rho}(S) := \frac{1}{|T|} \|Y_T - k(X_T, X_S) \tilde{A}_{S,\rho}^{-1} Y_S\|_2^2.$$

Then, for each player $i \in N$,

$$|\phi_i(U_{\infty,\rho}) - \phi_i(\tilde{U}_{\infty,\rho})| \leq \frac{4k_{\max}^2}{\rho^2} \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{1 + \log n}{n} \varepsilon_N \quad (7)$$

where $\varepsilon_N := \|\tilde{k}(X_N, X_N) - k(X_N, X_N)\|_2$.

For the theoretical analysis in Proposition 4.2, we assume that the test kernel matrix remains fixed for computational convenience, following the setting of (Cortes et al., 2010). The proposition leaves the construction of the approximate training kernel matrix $\tilde{k}(X_N, X_N)$ open, which allows us to compare different ways of constructing low-rank kernel approximations under the same stability bound.

Appendix E.5 compares the best rank- r approximation with the Nyström approximation. This comparison motivates

the eigenvalue-based approximation used in Algorithm 1: the best rank- r approximation is controlled directly by the spectral tail of the kernel matrix, whereas the Nyström approximation introduces an additional sampling-dependent error term. We also discuss how the required rank scales when the kernel eigenvalues obey a power-law decay.

5. Experiments

In this section, we demonstrate our algorithm works well compared to the baseline FreeShap (Wang et al., 2024a) in the small data selection. Specifically, we demonstrate that LRFShap achieve comparable to the baseline with significant speedup and scalable to huge datasets. Moreover, we explain cross-variation in selecting ability of our low-rank method using spectral alignment of both the kernel and the labels. Bordelon et al. (2020) also uses this term, but we apply it to data selection. All experimental results are averaged across 3 seeds unless stated.

5.1. Experimental setup

We evaluate data selection performance in the small data selection regime. We first approximate Shapley values with Algorithm 1 for LRFShap and Algorithm 2 for FreeShap. Then we select samples with the largest Shapley values and evaluate the prediction accuracy using the selected samples. Prediction is conducted by Equation 3 both in the baseline and LRFShap, to accurately compare the data selection ability of LRFShap compared to FreeShap. We run the algorithms with 500 TMC iterations. The datasets used are AG-News, MNLI, MR, MRPC, QQP, RTE, SST-2 (Socher et al., 2013; Williams et al., 2018; Dolan & Brockett, 2005; Giampiccolo et al., 2007; Pang & Lee, 2005; Zhang et al., 2015; Wang et al., 2019). The number of sample in each dataset is $\{1000, 2000, 5000\}$ for AG-News, MNLI, MR, QQP, and SST-2, and $\{1000, 2000, 3668\}$ for MRPC and $\{1000, 2000, 2490\}$ for RTE. The number of validation set is $\max(1000, |V_D|)$ for each dataset.

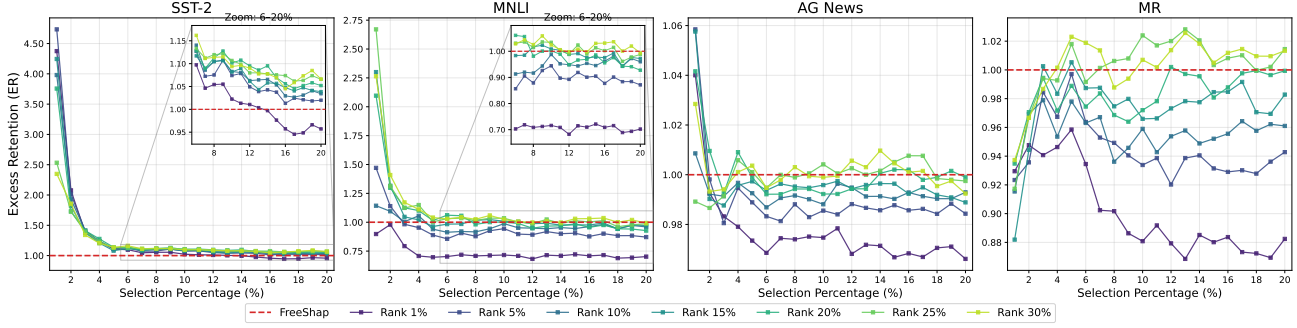


Figure 2. Excess Retention (ER) across selection percentages for four datasets. Note that in SST-2 and MNLI, our method can select significantly valuable data than the baseline when extremely small number of data are selected.

5.2. Accuracy preservation

Figure 1 represents the prediction accuracy when the 1 to 20% of the samples with the top Shapley values are selected, and Table 1 in Appendix B shows the absolute accuracy across ranks when 20% of the samples are selected. We measure how much our method selects valuable samples, we consider Excess Retention (ER) similar as in (Mnih et al., 2015) which is defined as

$$ER(r, p) = \frac{Acc_r(p) - Acc_{base}(p)}{Acc_{baseline}(p) - Acc_{base}}$$

where $Acc_r(p)$ is the accuracy of our method with the rank $r\%$, $Acc_{base}(p)$ is the accuracy of the baseline when $p\%$ of the data are selected for both, and Acc_{base} is the accuracy where none of the samples are selected. In Figure 2, we can see that moderately high rank preserves the excess retention well, indicating that with these ranks, we can effectively select valuable data samples. Meanwhile, in a few datasets with extremely small data selection (e.g. 1%), Figure 2 says that our method can select valuable data than the baseline significantly better. This suggests that the low-rank approximation may act as a spectral regularizer: when the label signal is sufficiently aligned with the leading eigenspace, truncating tail directions can preserve the useful ranking information while reducing unstable sample-specific components in TMC marginal gains (see Section 5.5 for relevant analyses). For the different affects of eigen ranks across datasets, see Section 5.5. For the results for other datasets and when the samples are selected across 1-100%, see Figures 7 to 10 in Appendix B.

5.3. Speedup by low-rank approximation

Figure 3 shows the speedup across ranks when the dataset size is 5000. In the figure, we can see that low ranks achieve significant speedup compared to the baseline, around $80\times$ for SST-2 $100\times$ for QQP. However, as the rank increases, the speedup decreases, with the minimum speedup about $4.1\times$. To see whether the speedup amplifies as the dataset size increases, see Section 5.4.

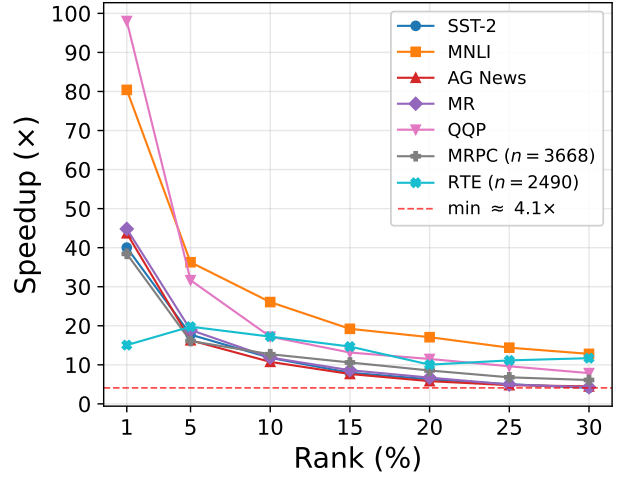


Figure 3. Speedups across the ranks at $n = 5000$ for seven datasets. RTE shows a different speedup trend, which is related to the early-stop ratio (see Figure 12).

Moreover, we see that low-rank approximation affects the early termination in TMC iteration. For example, if the predefined TMC iteration number is 500 and the early stop ratio is 10%, then TMC terminated at 50 iteration in average. Figure 4 says that in most of the case, lower rank makes TMC iterations terminate early, while for higher ranks TMC iterations stop later than the baseline. As mentioned in the baseline, we guess that the low rank approximation decreases the representation of the eNTK, saturating marginal contribution of a new sample early. This implies that if we choose a proper rank, we can more amplify the speedup by early stopping.

5.4. Speedup-accuracy trade-off

A next natural question after seeing the accuracy preservation and the speedup is whether this speedup is scalable. Figure 5 shows the speedup-accuracy trade-off across dataset sizes. For ranks ($\leq 10\%$), the Pareto frontier is improved as

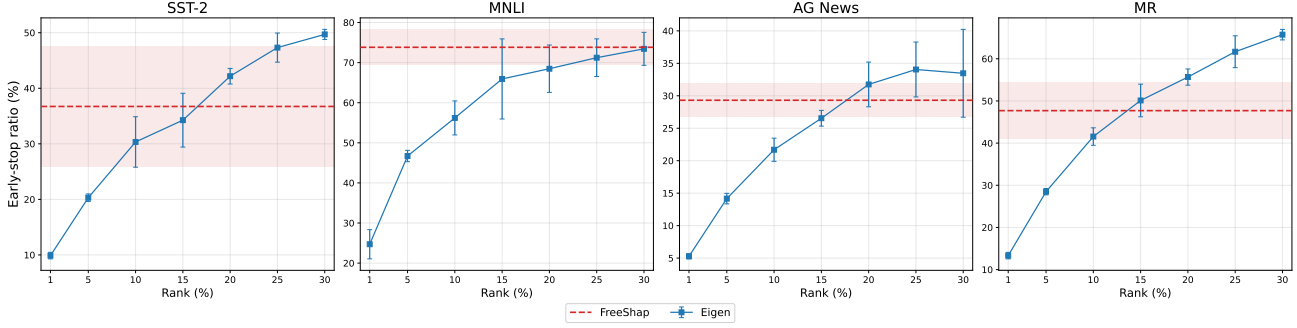


Figure 4. Average early stop ratio for FreeShap and LRFShap when $n = 5000$, with the shaded ± 1 standard deviation band (FreeShap) and the error bars (ours).

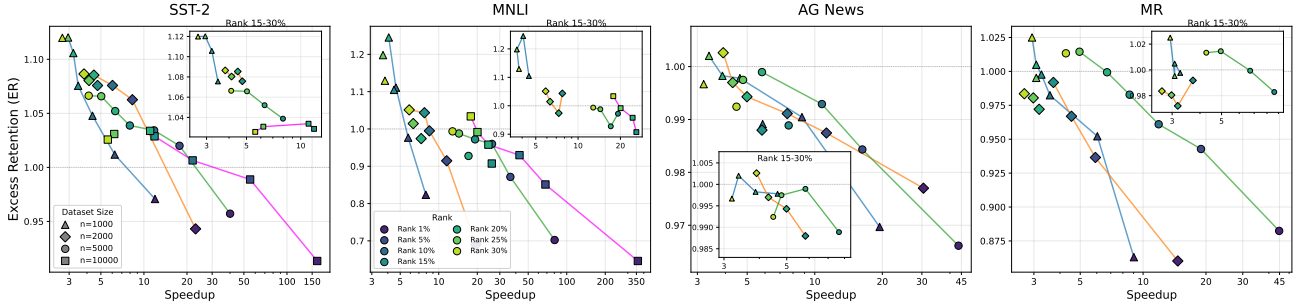


Figure 5. A Pareto frontier graphs for four datasets with the dataset sizes 1000, 2000, and 5000. For SST-2 and MNLI, the results with $n = 10000$ are also reported. Points with the rank 15–30% are represented in zoom panels for visual clarity. Due to the computation budget, the experiment for MNLI with $n = 10000$ was conducted with only one seed.

the dataset size increases: one can obtain the same ER with more speedup in larger datasets. For example, we can achieve $ER \approx 1$ about $20\times$ faster than the baseline when the dataset size is 10000 in SST-2 and MNLI. Moreover, when the dataset size is 5000, we can achieve ER equal or more than 0.95 with about $15 \sim 45\times$ speedups for all datasets. Considering Table 1, one can know that the actual accuracy ratio over the baseline is greater than 95% when $ER \approx 0.95$ because of the definition of ER. This implies that our method preserves accuracy well in real-world scenarios. However, in higher ranks, the frontier improvement is not consistently observed. We guess this is because the accuracy preservation is saturate, i.e., $ER \approx 1$ in higher ranks. One might think that the Pareto frontier is not clearly improved when the dataset size is changed from 1000 to 2000. We guess that this is because the difference of dataset size is relatively small. The improvement is clear when the dataset size becomes to 5000 and 10000. We expect that our scheme is scalable in selecting valuable data in huge datasets in real-world scenarios.

5.5. Spectral alignment analysis of dataset-dependent low-rank effects

In the above subsections, we saw that the accuracy preservation, i.e., excess retention (ER) is affected differently across

datasets. Since eigendecomposition is applied in TMC iteration in Algorithm 1, ER is connected to the data selection ability. In this section, we analyze the cause of this differentiation.

To analyze this phenomenon, we define two notations which we call *feature concentration* (FC) and *label concentration* (LC) which are used in (Bordelon et al., 2020; Canatar et al., 2021) as different terms. For the concise explanation, we consider the binary classification. As in Section 4, consider a training set D_N with labels $y_i \in \mathcal{Y}$ where $y_i \in \{0, 1\}$ for $i = 1, \dots, |N|$. Consider the eigendecomposition of the eNTK gram matrix $k(X_N, X_N)$: $k(X_N, X_N) = \sum_{i=1}^{|N|} \lambda_i u_i u_i^\top$ where λ_i 's are the eigenvalues with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{|N|} \geq 0$ and u_i 's the orthonormal eigenvectors corresponding to λ_i 's. Rank r decomposition of $k(X_N, X_N)$ can be expressed as $K_r = \sum_{i=1}^r \lambda_i u_i u_i^\top$ and $P_r := \sum_{i=1}^r u_i u_i^\top$ is the orthogonal projector onto the top- r eigenspace of $k(X_N, X_N)$. FC and LC with the rank r are defined as follows:

$$FC(r) := \frac{\sum_{i=1}^r \lambda_i}{\sum_{i=1}^{|S|} \lambda_i}, LC(r) := \frac{\sum_{i=1}^r (u_i^\top y)^2}{\|y\|_2^2} = \frac{\|P_r y\|_2^2}{\|y\|_2^2}.$$

A natural conjecture is through FC, since it directly says how

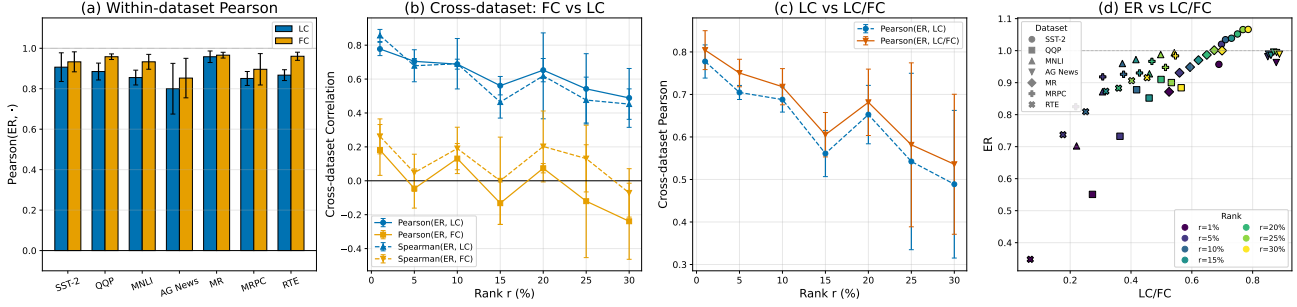


Figure 6. Pearson and Spearman correlations between ER and the various metrics FC, LC and LC/FC. (a) Pearson correlations within each dataset across ranks. (b) Pearson and Spearman correlations across datasets for a fixed rank. (c) The same cross-dataset correlations with (b) for LC and LC/FC. (d) Scatter plot of ER against LC/FC over all (dataset, rank) pairs. The error bars represent standard deviations. The missing Spearman correlations are shown in Appendix C.

much the representative information is included in K_r . We analyze two situation: ER across the rank r within a dataset and ER for a fixed r across datasets, both with Pearson and Spearman correlations. The results are shown in Figure 6. In Figure 6. (a), as expected, both FC and LC explains ER within a dataset well. One can see that FC gives a bit higher Pearson correlations than LC for all datasets. Also, FC and LC result in the same Spearman correlations, which is deferred to Appendix C.

However, Figure 6 (b) says that for a fixed rank r , FC does not explain ER across datasets. To find better explanation, we note the eNTK error from our low rank approximation in TMC iteration of Algorithm 1. By a standard spectral identity (Bach, 2013; Cortes et al., 2010) (proof in Appendix A), the in-sample disagreement between the full-NTK and rank- r LRNTK predictors on the training inputs satisfies

$$\|f_\rho^{\text{FNTK}}(X_N) - f_\rho^{\text{LRNTK}}(X_N)\|_2^2 \quad (8)$$

$$= \sum_{i>r} \left(\frac{\lambda_i}{\lambda_i + \rho} \right)^2 (\mathbf{u}_i^\top Y_N)^2 \leq (1 - \text{LC}(r)) \|Y_N\|_2^2. \quad (9)$$

The label-side quantity $1 - \text{LC}(r)$ thus provides a label-aware upper bound on the in-sample predictor gap, complementing the purely kernel-side bound $\|K_r - K\|_2$. Equation 9 suggest that LC helps quantify how much task-relevant label signal is lost by spectral truncation. Thus, unlike FC, LC provides a task-aware explanation for why the same rank can affect ER differently across datasets. Figure 6 (b) shows that both Pearson and Spearman correlations between ER and LC across datasets when the rank is fixed is significantly higher than that of FC as we guessed in above. This implies that in our method, geometric projection to top- r eigenspace of labels significantly affect to selecting valuable data. Note that in higher ranks, the correlations decreases in Figure 6 (b) and (c) because of the saturation of ER in higher ranks.

Moreover, we combine LC and FC to explain ER across datasets more accurately. As in Figure 6 (c), the Pearson cor-

relations between ER and LC/FC are a bit, but consistently higher than that of LC alone, indicating that LC/FC exhibits a slightly stronger linear relationship with ER across dataset than LC alone. The Spearman correlations for LC/FC are similar with those of LC, which are deferred to Appendix C. These results indicate that LC/FC, which computes the ratio of label alignment over the retained spectrum, explains data selection ability across datasets well. In addition, all of the correlations are higher in the lower ranks than the higher ranks since ER saturates in higher ranks. Figure 6 (d) plots ER against LC/FC for every (dataset, rank) pair. At each fixed rank, the points across datasets line up along a near-linear trend, visualizing the high cross-dataset Pearson and Spearman correlations reported in Figure 6 (c).

6. Conclusion

In this work, we further accelerated the eNTK based Shapley value approximation, and demonstrated that our method scalable, that is, we can achieve the same accuracy with more higher speedup when the dataset size increases. Moreover, we propose theoretical understanding to ensure that our method approximates Shapley well when dataset size is bigger, as in real-world scenarios. Finally, we give quantitative analysis by proposing factors that explain why low rank approximation affects the ability to select valuable samples across datasets.

We think interesting future research can be as follows: (i) extending existing work with Owen value which is an extension of Shapley value; (ii) online methods where valuable unseen data can selected with Shapley value. These research can contribute the efficient training of large language models and the interpretability of AI models.

Impact Statement

This work focuses on scalable data valuation and data selection. Our method may help practitioners identify useful data

more efficiently in large-scale settings. Since data selection can reflect biases in the data or in the chosen utility function, practical deployment should be accompanied by appropriate fairness, privacy, and domain-specific checks.

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A. Derivation of the predictor-gap bound equation 9

The bound equation 9 follows from the standard spectral analysis of kernel ridge regression and is essentially a restatement of the Nyström / spectral-truncation error bounds in (Bach, 2013; Cortes et al., 2010); we include the short derivation here for completeness.

Applying the push-through identity $\Phi^\top (\Phi \Phi^\top + \rho I)^{-1} = (\Phi^\top \Phi + \rho I)^{-1} \Phi^\top$ to $\Phi = \Phi_{\text{tr}}$ rewrites the LRNTK predictor as $f_\rho^{\text{LRNTK}}(X_N) = K_r (K_r + \rho I)^{-1} Y_N$ with $K_r := \Phi_{\text{tr}} \Phi_{\text{tr}}^\top$. Since K and K_r share the eigenbasis $\{\mathbf{u}_i\}$ —with eigenvalues λ_i for K and $\lambda_i \mathbb{1}_{[i \leq r]}$ for K_r —the spectral filters are

$$K(K + \rho I)^{-1} = \sum_{i=1}^n \frac{\lambda_i}{\lambda_i + \rho} \mathbf{u}_i \mathbf{u}_i^\top, \quad K_r(K_r + \rho I)^{-1} = \sum_{i \leq r} \frac{\lambda_i}{\lambda_i + \rho} \mathbf{u}_i \mathbf{u}_i^\top,$$

so their difference is supported entirely on the tail: $\sum_{i > r} \frac{\lambda_i}{\lambda_i + \rho} \mathbf{u}_i \mathbf{u}_i^\top$. Expanding $Y_N = \sum_i (\mathbf{u}_i^\top Y_N) \mathbf{u}_i$ and using orthonormality gives the equality in equation 9. The inequality follows from $\lambda_i / (\lambda_i + \rho) \in [0, 1)$ for $\rho > 0$ together with $\sum_{i > r} (\mathbf{u}_i^\top Y_N)^2 = (1 - \text{LC}(r)) \|Y_N\|_2^2$, and is tight as $\rho \rightarrow 0^+$.

B. Figures and Table omitted in the main paper

In this section, we report the figures and a table which are omitted in the main paper due to the page limit.

Table 1. Accuracy at 20% selection, mean over 3 seeds. FreeShap is the baseline; subsequent columns are Eigen decomposition with the rank $r\%$ in our method. $n=5000$ unless noted. Base means when no data was selected, i.e., a pre-trained model.

Dataset	FreeShap	rank							
		Base	1%	5%	10%	15%	20%	25%	30%
SST-2	0.8391	4908	0.8230	0.8448	0.8498	0.8517	0.8563	0.8613	0.8612
MNLI	0.6420	0.3640	0.5587	0.6063	0.6307	0.6333	0.6207	0.6383	0.6400
AG News	0.8967	0.2370	0.8743	0.8863	0.8920	0.8893	0.8960	0.8950	0.8917
MR	0.8537	0.5047	0.8127	0.8337	0.8400	0.8477	0.8533	0.8587	0.8583
QQP	0.8033	0.6137	0.7187	0.7530	0.7803	0.7753	0.7860	0.7843	0.7813
MRPC ($n=3668$)	0.7647	0.3162	0.6830	0.7247	0.7279	0.7295	0.7459	0.7377	0.7541
RTE ($n=2490$)	0.8628	0.5271	0.6450	0.7762	0.8003	0.8219	0.8255	0.8327	0.8363

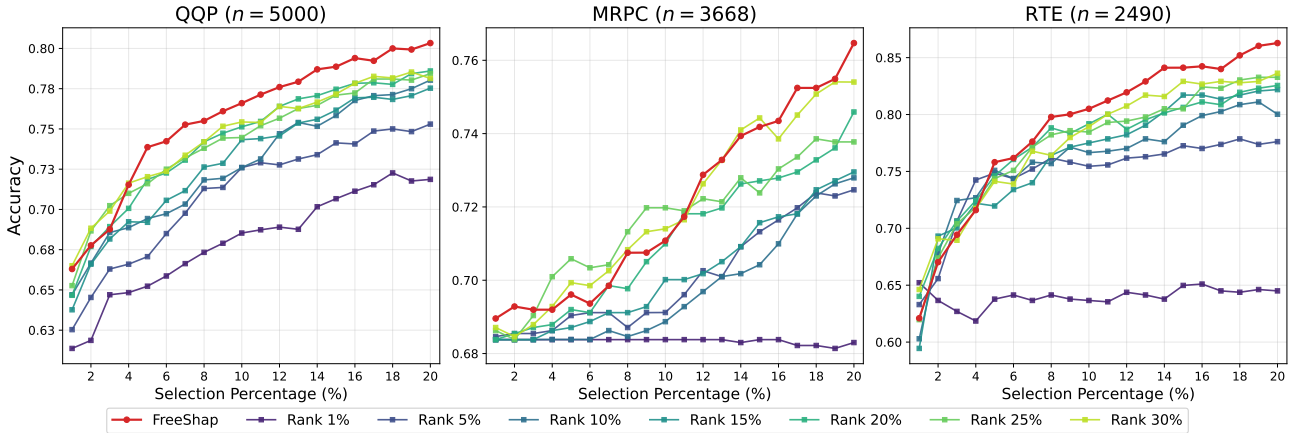


Figure 7. Accuracy per selected dataset for three datasets, QQP, MRPC, and RTE across 1-20% of the train data.

C. Missing Spearman correlations in Section 5

Figure 13 reports the omitted Spearman correlations in Figure 6. In this figure, one can see that Spearman correlations are similar in all of the graphs. Also the scatter graphs demonstrate that FC alone exhibits no clear monotone alignment with

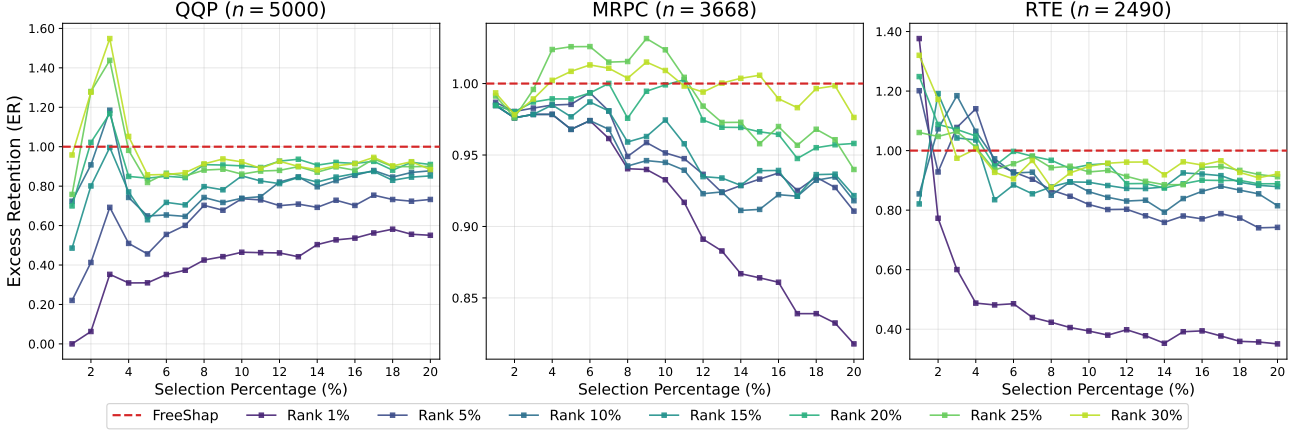


Figure 8. ER for QQP, MRPC, RTE across 1-20% of the train data.

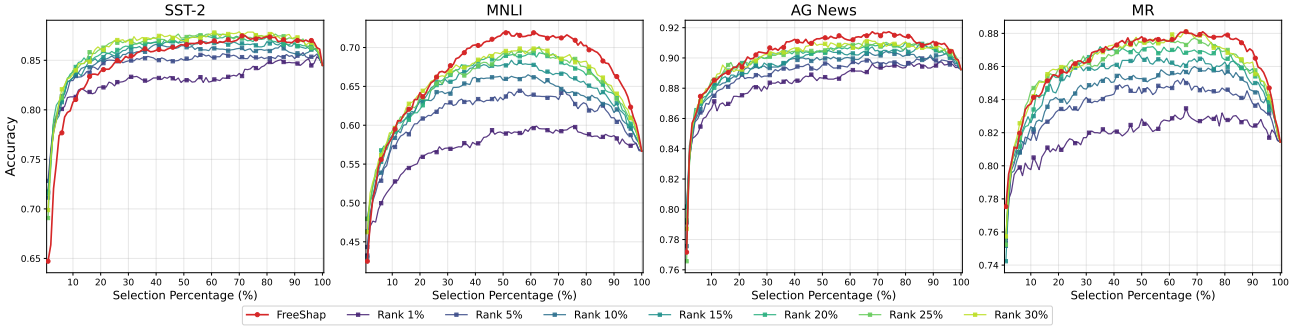


Figure 9. Accuracy per selected dataset for SST-2, MNLI, AG News, and MR across 1-100% of the train data. One can see that the data selection ability of our method becomes relatively low when the number of selected becomes larger.

RE, and LC/FC yields the tightest near-linear cross-task alignment, confirming that the ratio—rather than either component alone—is the most reliable a priori predictor of ER.

D. Detailed preliminaries

D.1. Data Shapley

We restrict our attention to the supervised learning setting. Given a dataset $D_N \cup D_T$, we consider the empirical risk minimizer

$$f_S := \arg \inf_{f \in \mathcal{F}} \left\{ \frac{1}{|S|} \sum_{(x_i, y_i) \in D_S} l(f(x_i), y_i) \right\},$$

where $\mathcal{F}$ is a hypothesis space corresponding to a chosen model class, $D_S \subseteq D_N$ is a subset of the training data, and f_S denotes the model trained on D_S .

We define a utility function $U(S)$ that measures the contribution of the subset S by evaluating the performance of f_S on the test set D_T . Typical choices of utility functions include test accuracy and test error. In particular, we consider the following two utility functions:

$$U_{\text{acc}}(S) := \frac{1}{|T|} \sum_{(x_j, y_j) \in D_T} \mathbf{1}[f_S(x_j) = y_j], \quad (10)$$

$$U_{\text{err}}(S) := \frac{1}{|T|} \sum_{(x_j, y_j) \in D_T} (f_S(x_j) - y_j)^2. \quad (11)$$

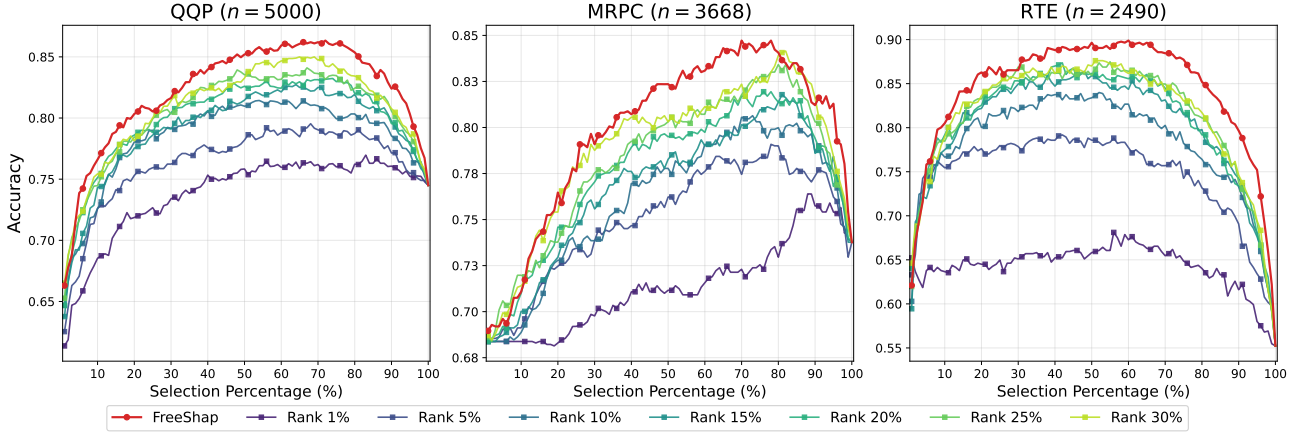


Figure 10. Accuracy per selected dataset for SST-2, MNLI, AG News, and MR across 1-100% of the train data. One can see that the data selection ability of our method becomes relatively low when the number of selected becomes larger.

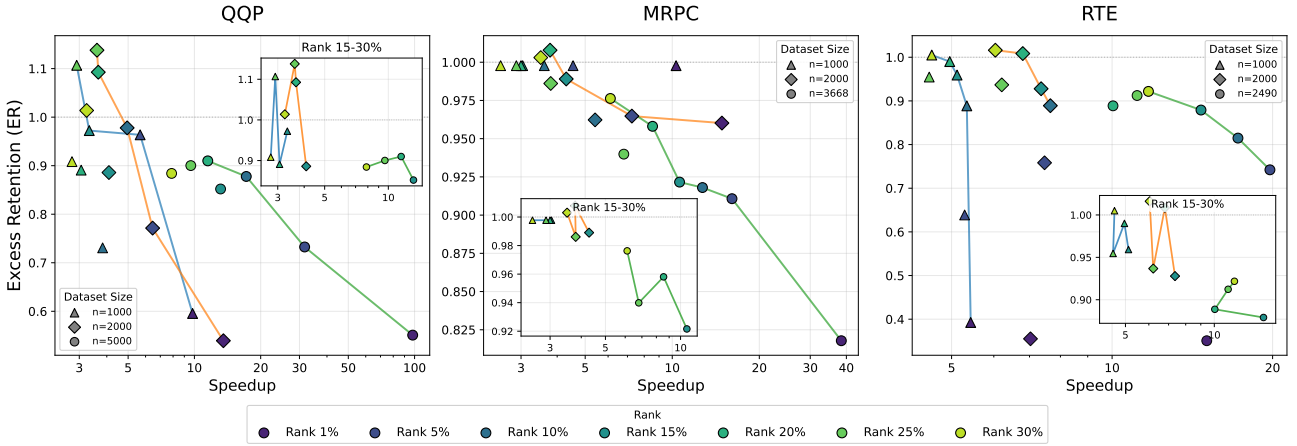


Figure 11. Pareto frontier graphs for QQP, MRPC, RTE with various dataset sizes. These datasets illustrate different results from Figure 5. QQP shows the similar result, MRPC for $n = 1000$ shows the almost same ER across ranks since a model trained with MRPC too small data does not increase accuracy even with the baseline Wang et al. (2024a). The Pareto frontiers for RTE have different shapes compared to other datasets. This is because for RTE, early stop ratio does not consistently increase as rank increases (see Figure 12).

We now formulate the data valuation problem as a cooperative game by treating each data point as a player and using the utility function $U(S)$ defined above. The Shapley value assigns to each data point a value that quantifies its contribution, and this approach is commonly referred to as *Data Shapley* (Ghorbani & Zou, 2019; Jia et al., 2019; Wu et al., 2023).

However, as can be seen from Equation 1, computing the Shapley value exactly requires exponential time in the number of training samples. Moreover, for each subset $S \subseteq N$, evaluating the marginal contribution $U(S \cup \{i\}) - U(S)$ typically requires training (or retraining) a model f_S on the subset S .

Therefore, the Data Shapley framework faces two major computational challenges:

1. **Combinatorial complexity:** The intrinsic exponential cost of computing the Shapley value itself.
2. **Model retraining cost:** The need to retrain a separate model f_S for each subset S in order to evaluate marginal contributions.

Regarding the first challenge, a well-known approximation method is Monte Carlo (MC) sampling based on the permutation formulation in Equation 1. MC sampling provides an unbiased estimator of the Shapley value, and error bounds based on Hoeffding’s inequality were established in (Maleki et al., 2013). More advanced approaches include the group testing-based method (Jia et al., 2019) and the stratified sampling method (Wu et al., 2023).

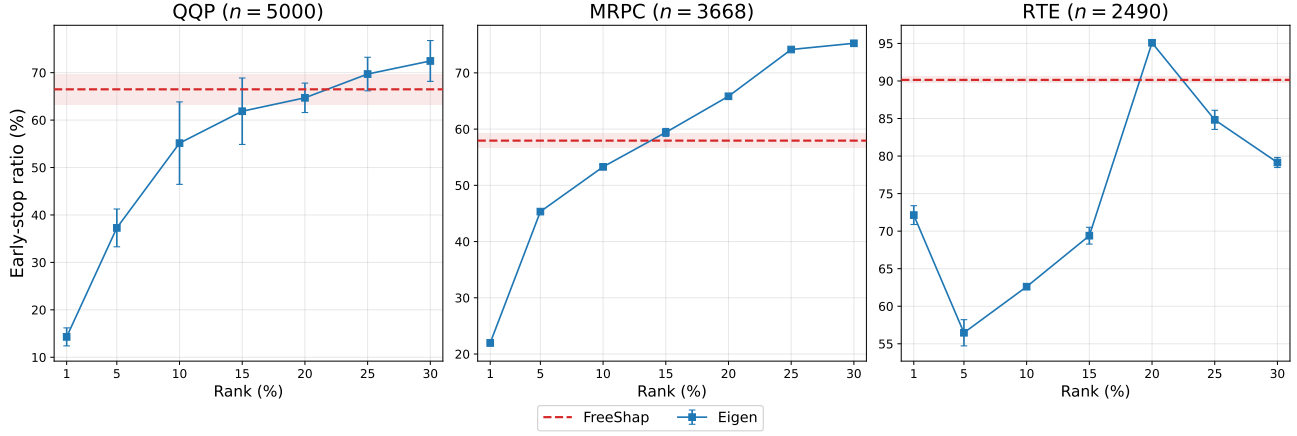


Figure 12. Average early stop ratio for QQP, MRPC, and RTE. RTE does not increasing tendency of the ratio along the rank. It seems that RTE has different expressiveness across the ranks.

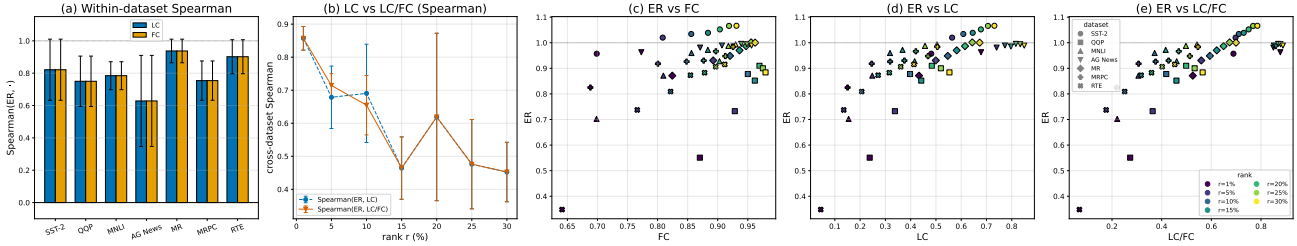


Figure 13. Missing Spearman correlations and scatter figures in Figure 6. (a) Spearman correlations within each dataset. (b) Spearman Correlations across datasets for each fixed rank. (c), (d), (e) Scatter plots of ER against FC, LC, and LC/FC over all (dataset, rank) pairs.

Concerning the second challenge, (Wang et al., 2024a) proposed replacing explicit model retraining for each subset S with a kernel regression procedure based on a precomputed empirical Neural Tangent Kernel (eNTK). This fine-tuning-free approach significantly reduces the computational burden.

D.2. Neural Tangent Kernel

We consider a parametric model $f(x; \theta) : \mathbb{R}^d \rightarrow \mathbb{R}^C$ that is differentiable with respect to the parameter vector $\theta \in \mathbb{R}^P$. Let $\theta_0 \in \mathbb{R}^P$ denote an initialization of the model parameters. The empirical neural tangent kernel (eNTK) associated with the model at θ_0 is defined, for any pair of inputs $x_1, x_2 \in \mathbb{R}^d$, as

$$\Theta_{\theta_0}(x_1, x_2) := \underbrace{\left[\frac{\partial f(x_1; \theta_0)}{\partial \theta_0} \right]_{C \times P}}_{C \times P} \underbrace{\left[\frac{\partial f(x_2; \theta_0)}{\partial \theta_0} \right]_{P \times C}^T}_{P \times C} \in \mathbb{R}^{C \times C}. \quad (12)$$

Here, $\frac{\partial f(x; \theta_0)}{\partial \theta_0} \in \mathbb{R}^{C \times P}$ denotes the Jacobian of the model output with respect to the flattened parameter vector θ_0 .

In the *kernel regime*, the network parameters remain sufficiently close to their initialization throughout training. Consequently, the network dynamics are well approximated by the first-order Taylor expansion around θ_0 , namely,

$$f(x; \theta_t) \approx f(x; \theta_0) + \frac{\partial f(x; \theta_0)}{\partial \theta_0} (\theta_t - \theta_0), \quad \forall t \geq 0. \quad (13)$$

Moreover, the empirical neural tangent kernel remains approximately constant along the training trajectory,

$$\Theta_{\theta_t}(x_1, x_2) \approx \Theta_{\theta_0}(x_1, x_2), \quad \forall x_1, x_2 \in \mathbb{R}^d. \quad (14)$$

As a result, the evolution of the network outputs is governed by a fixed kernel determined at initialization.

Under this approximation, training the neural network with gradient flow reduces to performing kernel gradient flow in function space. In particular, when the squared loss is used without explicit regularization, the resulting dynamics coincide with the gradient flow of *ridgeless kernel regression* associated with the eNTK Θ_{θ_0} .

Let the training dataset be denoted by D_S with index set $S := \{1, \dots, |S|\}$. We define the input matrix as $X_S := [x_1, \dots, x_{|S|}]^\top$ and the corresponding one-hot encoded label matrix as $Y_S := [y_1, \dots, y_{|S|}]^\top \in \{0, 1\}^{|S| \times C}$. We further use $\text{vec}(Y_S) \in \{0, 1\}^{|S|C}$ to denote the vectorized form of the label matrix.

Given a test input x_t , the prediction of the eNTK regression model is given by

$$f_S^{\text{eNTK}}(x_t) = \underbrace{f(x_t; \theta_0)}_{C \times 1} + \underbrace{K(x_t, X_S)^\top}_{C \times |S|C} \underbrace{K(X_S, X_S)^\dagger}_{|S|C \times |S|C} \underbrace{\text{vec}(Y_S - f(X_S; \theta_0))}_{|S|C \times 1}, \quad (15)$$

where $K(x_t, X_S) = [\Theta_{\theta_0}(x_t, x_j)]_{j=1}^{|S|} \in \mathbb{R}^{|S|C \times C}$ and $K(X_S, X_S) = [\Theta_{\theta_0}(x_i, x_j)]_{i,j=1}^{|S|} \in \mathbb{R}^{|S|C \times |S|C}$.

We make two remarks regarding the practical use of the empirical neural tangent kernel.

First, one may ask whether it is possible to use pretrained weights, rather than random initialization, as the reference parameter θ_0 for defining the empirical neural tangent kernel. In the context of natural language processing, this question was investigated by (Malladi et al., 2023), while related results for computer vision tasks were reported in (Wei et al., 2022).

Second, directly inverting the kernel matrix $K(X_S, X_S)$ in Equation 15 requires $\mathcal{O}((|S|C)^3)$ computational cost, which becomes prohibitive when the number of classes C is large. To mitigate this class-dependent inefficiency, various approximation techniques have been proposed, including the first-logit approximation.

From a theoretical perspective, (Mohamadi et al., 2023) provides a principled justification for such approximations. Specifically, they consider a multi-output model $f(x; \theta) = [f^1(x; \theta), \dots, f^C(x; \theta)]$ and introduces a pseudo neural tangent kernel (pNTK) defined by

$$\hat{\Theta}_{\theta_0}(x_1, x_2) := \underbrace{\left[\frac{\partial}{\partial \theta_0} \left(\frac{1}{\sqrt{C}} \sum_{i=1}^C f^i(x_1; \theta_0) \right) \right]}_{1 \times P} \underbrace{\left[\frac{\partial}{\partial \theta_0} \left(\frac{1}{\sqrt{C}} \sum_{i=1}^C f^i(x_2; \theta_0) \right) \right]^\top}_{P \times 1}. \quad (16)$$

Using this kernel, they show that the regression predictor in Equation 15 admits the following closed-form approximation:

$$f_S^{\text{pNTK}}(x_t) = \underbrace{f(x_t; \theta_0)}_{1 \times C} + \underbrace{\hat{K}(x_t, X_S)^\top}_{1 \times |S|} \underbrace{\hat{K}(X_S, X_S)^\dagger}_{|S| \times |S|} \underbrace{(Y_S - f(X_S; \theta_0))}_{|S| \times C}, \quad (17)$$

where $\hat{K}(x_t, X_S) = [\hat{\Theta}_{\theta_0}(x_t, x_j)]_{j=1}^{|S|} \in \mathbb{R}^{|S|}$ and $\hat{K}(X_S, X_S) = [\hat{\Theta}_{\theta_0}(x_i, x_j)]_{i,j=1}^{|S|} \in \mathbb{R}^{|S| \times |S|}$.

Importantly, this approximation has been shown empirically to remain effective when θ_0 is chosen as pretrained weights, rather than random initialization. To facilitate subsequent analysis, we rewrite Equation 17 in vectorized form over a test set X_T and introduce an explicit ridge regularization parameter $\rho > 0$. This yields

$$F_S^{\text{pNTK}}(X_T) = \underbrace{f(X_T; \theta_0)}_{|T| \times C} + \underbrace{\hat{K}(X_T, X_S)^\top}_{|T| \times |S|} \underbrace{(\hat{K}(X_S, X_S) + \rho I_{|S|})^{-1}}_{|S| \times |S|} \underbrace{(Y_S - f(X_S; \theta_0))}_{|S| \times C}. \quad (18)$$

Define the residual targets $\tilde{Y}_S := Y_S - f(X_S; \theta_0)$ and the residual predictor $\tilde{F}_S^{\text{pNTK}}(X_T) := F_S^{\text{pNTK}}(X_T) - f(X_T; \theta_0)$. Then Equation 18 can be equivalently expressed as

$$\tilde{F}_S^{\text{pNTK}}(X_T) = k(X_T, X_S)^\top (k(X_S, X_S) + \rho I_{|S|})^{-1} \tilde{Y}_S, \quad (19)$$

where we write $k = \hat{K}$ for simplicity.

D.3. FreeShap algorithm

We report our baseline algorithm (Wang et al., 2024a).

Algorithm 2 FreeShap

Require: Training set D_N , test set D_T , utility function U , maximum MC iterations M , eNTK K , $\rho > 0$

Ensure: Shapley values $\phi_1, \dots, \phi_n$

- 1: Initialize $\phi_i \leftarrow 0$ for $i = 1, \dots, n$
- 2: Load precomputed eNTK matrices $K_{tr} \leftarrow k(X_N, X_N)$, $K_{te} \leftarrow k(X_T, X_N)$
- 3: **for** $t = 1$ to M **do**
- 4: Sample a permutation π^t of N
- 5: $u_0^t \leftarrow U(\emptyset, D_T)$
- 6: **for** $p = 1$ to n **do**
- 7: $S_p^t \leftarrow \{\pi^t[1], \dots, \pi^t[p]\}$
- 8: Extract submatrix $k(X_{S_p^t}, X_{S_p^t})$ and $k(X_T, X_{S_p^t})$
- 9: Compute ridge predictor $f_{S_p^t}^{eNTK}(X_T) \leftarrow k(X_T, X_{S_p^t}) \left(k(X_{S_p^t}, X_{S_p^t}) + \rho I_{S_p^t} \right)^{-1} Y_{S_p^t}$ **{Bottleneck.}**
- 10: $u_p^t \leftarrow U(S_p^t, D_T | f_{S_p^t}^{eNTK})$
- 11: Update Shapley estimate $\phi_{\pi^t[p]} \leftarrow \frac{t-1}{t} \phi_{\pi^t[p]} + \frac{1}{t} (u_p^t - u_{p-1}^t)$
- 12: **end for**
- 13: **end for**

E. Theoretical understanding

E.1. Solutions by kernel regression

Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a positive semi-definite kernel function, and let $\mathcal{H}_k$ denote the associated reproducing kernel Hilbert space (RKHS). Given a training dataset $D_S = \{(x_i, y_i)\}_{i=1}^{|S|}$ with $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, we consider the following empirical risk functional:

$$\mathcal{L}_\rho^S(f) := \frac{1}{|S|} \sum_{i=1}^{|S|} (f(x_i) - y_i)^2 + \rho \|f\|_{\mathcal{H}_k}^2 \quad (20)$$

where $\rho \geq 0$ is a regularization parameter.

Let $f_{t,\rho}^S$ denote the function obtained by running gradient flow on the regularized objective $\mathcal{L}_\rho^S$ in the RKHS $\mathcal{H}_k$, initialized at $f_0 = 0$ and evolved up to time $t \in [0, \infty)$ with a fixed regularization parameter $\rho > 0$. According to Appendix C of (Haas et al., 2023), the solution admits the explicit representation

$$f_{t,\rho}^S(x) = k(x, X_S) \left(I_{|S|} - e^{-\frac{2}{|S|}t} \left(k(X_S, X_S) + \rho |S| I_{|S|} \right) \right) \left(k(X_S, X_S) + \rho |S| I_{|S|} \right)^{-1} Y_S, \quad (21)$$

where $k(x, X_S) := [k(x, x_i)]_{i \in S}$ and $k(X_S, X_S) := [k(x_i, x_j)]_{i,j \in S}$ denotes the kernel matrix.

This expression provides a unified perspective on several classical kernel estimators:

1. kernel ridge regression in the limit $t \rightarrow \infty$;
2. gradient flow on the unregularized objective as $\rho \downarrow 0$;
3. ridgeless kernel regression $f_{\infty,0}(x) = k(x, X_S) k(X_S, X_S)^\dagger y$ in the joint limit $\rho \rightarrow 0$ and $t \rightarrow \infty$.

E.2. Spectral analysis for the finite-time error of gradient flow

Our goal is to analyze the finite-time error $\|f_{\infty,\rho}^S - f_{t,\rho}^S\|_{\mathcal{H}_k}^2$ of the gradient flow solution for each subset $S \subseteq N$. To this end, we express $f_{t,\rho}^S(x) = k(x, X_S) \alpha_{t,\rho}^S$ where the coefficient vector $\alpha_{t,\rho}^S \in \mathbb{R}^{|S|}$ is given by

$$\alpha_{t,\rho}^S = \left(I_{|S|} - e^{-\frac{2}{|S|}t} \left(k(X_S, X_S) + \rho |S| I_{|S|} \right) \right) \left(k(X_S, X_S) + \rho |S| I_{|S|} \right)^{-1} Y_S. \quad (22)$$

Then, the limiting coefficient vector is $\alpha_{\infty,\rho}^S = (k(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S$, and the corresponding finite-time residual is $\Delta\alpha_\rho^S := \alpha_{t,\rho}^S - \alpha_{\infty,\rho}^S = -e^{-\frac{2}{|S|}t(k(X_S, X_S) + \rho|S|I_{|S|})}(k(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S$. From this, since $f_{\infty,\rho}^S - f_{t,\rho}^S = k(\cdot, X_S)\Delta\alpha_\rho^S$, the RKHS norm satisfies

$$\|f_{\infty,\rho}^S - f_{t,\rho}^S\|_{\mathcal{H}}^2 = (\Delta\alpha_\rho^S)^\top k(X_S, X_S)(\Delta\alpha_\rho^S).$$

Using the eigen decomposition we have $k(X_S, X_S) = V_S \Lambda_S V_S^\top$ where $\Lambda_S = \text{diag}(\lambda_1^S, \dots, \lambda_{|S|}^S)$ is diagonal and $V_S \in \mathbb{R}^{|S| \times |S|}$ is orthonormal. Then we can represent the residual as

$$\Delta\alpha_\rho^S = V_S e^{-\frac{2}{|S|}t(\Lambda_S + \rho|S|I_{|S|})}(\Lambda_S + \rho|S|I_{|S|})^{-1}V_S^\top Y_S.$$

Defining rotated coordinates $\tilde{Y}_S := V_S^\top Y_S$, a direct computation yields the exact spectral identity

$$\|f_{\infty,\rho}^S - f_{t,\rho}^S\|_{\mathcal{H}}^2 = (V_S^\top \Delta\alpha_\rho^S)^\top \Lambda_S (V_S^\top \Delta\alpha_\rho^S) = \sum_{i=1}^{|S|} \lambda_i^{|S|} (V_S^\top \Delta\alpha_\rho^S)_i^2 = \sum_{i=1}^{|S|} \lambda_i^S \left(\frac{e^{-\frac{2t}{|S|}(\lambda_i^S + \rho|S|)}}{\lambda_i^S + \rho|S|} \right)^2 (\tilde{Y}_{S,i})^2. \quad (23)$$

Case 1: $\rho = 0$ (ridgeless). When $\rho = 0$, only the strictly positive eigenvalues of $k(X_S, X_S)$ contribute:

$$\|f_{\infty,0}^S - f_{t,0}^S\|_{\mathcal{H}_k}^2 = \sum_{i:\lambda_i^S > 0} \frac{e^{-\frac{4t}{|S|}\lambda_i^S}}{\lambda_i^S} (\tilde{Y}_{S,i})^2.$$

Since $e^{-x} \leq \frac{1}{x}$ for $x > 0$, we have

$$\|f_{\infty,0}^S - f_{t,0}^S\|_{\mathcal{H}_k}^2 \leq \frac{|S|}{4t} \|k(X_S, X_S)^\dagger Y_S\|_2^2 \leq \frac{|S| \|Y_S\|_2^2}{4t(\lambda_{\min}^+(k(X_S, X_S)))^2}, \quad (24)$$

where $\lambda_{\min}^+(k(X_S, X_S)) := \min\{\lambda_i^S : \lambda_i^S > 0\}$.

Case 2: $\rho > 0$ (regularized) When $\rho > 0$, we have $\lambda_i^S + \rho|S| \geq \rho|S| > 0$. Hence from Equation 23, we have

$$\|f_{\infty,\rho}^S - f_{t,\rho}^S\|_{\mathcal{H}_k}^2 \leq e^{-4t\rho} \sum_{i=1}^{|S|} \frac{(\tilde{Y}_{S,i})^2}{\lambda_i^S + \rho|S|} = Y_S^\top (k(X_S, X_S) + \rho|S|I_{|S|})^{-1} Y_S \leq \frac{e^{-4t\rho}}{\rho|S|} \|Y_S\|_2^2. \quad (25)$$

E.3. Bounding the estimation error for Shapley value

Lemma E.1 (Size-dependent perturbation bound for Shapley value). *Let $N = \{1, \dots, n\}$ be the player set, and let $U, V : 2^N \rightarrow \mathbb{R}$ be two cooperative games. Assume that there exists a function $g : \{0, 1, \dots, n\} \rightarrow \mathbb{R}_{\geq 0}$ such that for every $l = 0, \dots, n$ and every $S \subseteq N$ with $|S| = l$, $|U(S) - V(S)| \leq g(l)$.*

Then for every $i \in N$,

$$|\phi_i(U) - \phi_i(V)| \leq \frac{1}{n} \left(g(0) + g(n) + 2 \sum_{l=1}^{n-1} g(l) \right).$$

The above result follows directly from the size-grouped representation of the Shapley value Equation 1 together with the additivity axiom of the Shapley value. Moreover, if $g(0) = 0$, we obtain the following simplified bound: $|\phi_i(U) - \phi_i(V)| \leq \frac{2}{n} \sum_{l=1}^n g(l)$. In particular, if the perturbation bound does not depend on the coalition size, that is, if there exists a constant $\varepsilon \geq 0$ such that $|U(S) - V(S)| \leq \varepsilon$ for all $S \subseteq N$, then taking $g(l) \equiv \varepsilon$ yields $|\phi_i(U) - \phi_i(V)| \leq 2\varepsilon$, which coincides with the bound obtained by directly controlling the supremum $\sup_{S \subseteq N} |U(S) - V(S)|$.

E.4. Proof of Proposition 1

For convenience, we restate Proposition 4.1.

Proposition E.1. Assume $\rho > 0$. Let D_N be a training dataset and D_T be a test dataset. For each subset $S \subseteq N$, define the utility function

$$U_{t,\rho}(S) := \frac{1}{|T|} \|Y_T - f_{t,\rho}^S(X_T)\|_2^2,$$

where $f_{t,\rho}^S$ denotes the gradient flow solution at time t with regularization ρ in Equation 5.

Suppose further that the following conditions hold:

(i) **(Bounded diagonal)** $\sup_x k(x, x) \leq k_{\max}^2$.

(ii) **(Bounded labels)** $|y_j| \leq 1$ for all j . In particular, this implies $\|Y_T\|_2 \leq \sqrt{|T|}$ and $\|Y_S\|_2 \leq \sqrt{|S|}$.

Then, for each player $i \in N$, the corresponding Shapley values satisfy

$$|\phi_i(U_{t,\rho}) - \phi_i(U_{\infty,\rho})| \leq \frac{4k_{\max}}{\sqrt{\rho}} \left(1 + \frac{k_{\max}}{\sqrt{\rho}}\right) e^{-2t\rho}.$$

Proof. Fix an arbitrary subset $S \subseteq N$. We begin by estimating the difference between the two utilities. Observe that

$$\begin{aligned} |U_{t,\rho}(S) - U_{\infty,\rho}(S)| &= \frac{1}{|T|} \left| \|Y_T - f_{t,\rho}^S(X_T)\|_2^2 - \|Y_T - f_{\infty,\rho}^S(X_T)\|_2^2 \right| \\ &\leq \frac{1}{|T|} \|2Y_T - f_{t,\rho}^S(X_T) - f_{\infty,\rho}^S(X_T)\|_2 \|f_{t,\rho}^S(X_T) - f_{\infty,\rho}^S(X_T)\|_2 \\ &\leq \frac{1}{|T|} (\|Y_T - f_{t,\rho}^S(X_T)\|_2 + \|Y_T - f_{\infty,\rho}^S(X_T)\|_2) \|f_{t,\rho}^S(X_T) - f_{\infty,\rho}^S(X_T)\|_2 \end{aligned}$$

By the reproducing property of the RKHS,

$$\|f(X_T)\|_2^2 = \sum_{x_i \in D_T} |f(x_i)|^2 \leq \|f\|_{\mathcal{H}_k}^2 \sum_{x_i \in D_T} k(x_i, x_i) = \|f\|_{\mathcal{H}_k}^2 \text{tr } k(X_T, X_T).$$

Moreover, by Proposition 2 in (Haas et al., 2023), for any $t \in [0, \infty]$ and $\rho \geq 0$, it holds that $\|f_{t,\rho}^S\|_{\mathcal{H}_k} \leq \|f_{\infty,\rho}^S\|_{\mathcal{H}_k} \leq \|f_{\infty,0}^S\|_{\mathcal{H}_k}$. Consequently, we obtain

$$\begin{aligned} &|U_{t,\rho}(S) - U_{\infty,\rho}(S)| \\ &\leq \frac{1}{|T|} (\|Y_T - f_{t,\rho}^S(X_T)\|_2 + \|Y_T - f_{\infty,\rho}^S(X_T)\|_2) \|f_{t,\rho}^S - f_{\infty,\rho}^S\|_{\mathcal{H}_k} \sqrt{\text{tr}(k(X_T, X_T))} \\ &\leq \frac{2}{|T|} \left(\sqrt{|T|} + \|f_{\infty,\rho}^S\|_{\mathcal{H}_k} \sqrt{\text{tr } k(X_T, X_T)} \right) \|f_{t,\rho}^S - f_{\infty,\rho}^S\|_{\mathcal{H}_k} \sqrt{\text{tr}(k(X_T, X_T))} \\ &\leq \frac{2}{\sqrt{|T|}} (1 + k_{\max} \|f_{\infty,\rho}^S\|_{\mathcal{H}_k}) \|f_{t,\rho}^S - f_{\infty,\rho}^S\|_{\mathcal{H}_k} \sqrt{\text{tr}(k(X_T, X_T))} \\ &\leq 2k_{\max} (1 + k_{\max} \|f_{\infty,\rho}^S\|_{\mathcal{H}_k}) \|f_{t,\rho}^S - f_{\infty,\rho}^S\|_{\mathcal{H}_k}. \end{aligned}$$

Since $f_{\infty,\rho}^S$ is the unique minimizer of the regularized risk Equation 20, comparing it with the zero function gives

$$\rho \|f_{\infty,\rho}^S\|_{\mathcal{H}_k}^2 \leq \mathcal{L}_\rho^S(f_{\infty,\rho}^S) \leq \mathcal{L}_\rho^S(0) = \frac{1}{|S|} \|Y_S\|_2^2 \leq 1.$$

Combining the above estimates and using the spectral decay bound Equation 25, we have

$$|U_{t,\rho}(S) - U_{\infty,\rho}(S)| \leq 2k_{\max} \left(1 + \frac{k_{\max}}{\sqrt{\rho}}\right) \sqrt{\frac{e^{-4t\rho}}{\rho}}.$$

Finally, the corresponding Shapley value difference follows immediately by applying Lemma E.1. $\square$

Remark E.1. In the special case $\rho = 0$, i.e., the ridgeless setting, the previous argument together with E.4 and Equation 24 yields the following estimate: for any $S \subset N$ and $t \in [0, \infty]$,

$$|U_{t,0}(S) - U_{\infty,0}(S)| \leq 2k_{\max}(1 + k_{\max}\|f_{\infty,0}^S\|_{\mathcal{H}_k}) \frac{|S|\|Y_S\|_2^2}{4t(\lambda_{\min}^+(k(X_S, X_S)))^2}.$$

However, even for principal submatrices of a positive semi-definite matrix, there is generally no uniform positive lower bound on the smallest nonzero eigenvalue. Consequently, since the computation of Shapley values involves all subsets $S \subset N$, it becomes difficult to establish a uniform stability bound for Data Shapley along the gradient flow in the ridgeless setting. This issue is also reflected in the fact that the bound in Equation 6 deteriorates and eventually blows up as $\rho \rightarrow 0$.

E.5. Proof of Proposition 2

Proposition E.2. Assume $\rho > 0$. Let D_N be a training dataset and D_T a test dataset. For each subset $S \subseteq N$, define the utility functions

$$U_{\infty,\rho}(S) := \frac{1}{|T|} \left\| Y_T - k(X_T, X_S)(k(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S \right\|_2^2,$$

and

$$\tilde{U}_{\infty,\rho}(S) := \frac{1}{|T|} \left\| Y_T - k(X_T, X_S)(\tilde{k}(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S \right\|_2^2.$$

Here, $\tilde{k}(X_N, X_N)$ is an approximation of the kernel matrix $k(X_N, X_N)$, and both matrices are assumed to be symmetric positive semi-definite.

Suppose that the following conditions hold:

- (i) **(Bounded diagonal)** $\sup_x k(x, x) \leq k_{\max}^2$.
- (ii) **(Bounded labels)** $|y_j| \leq 1$ for all j . In particular, $\|Y_T\|_2 \leq \sqrt{|T|}$ and $\|Y_S\|_2 \leq \sqrt{|S|}$.

Then, for each player $i \in N$, the corresponding Shapley values satisfy

$$|\phi_i(U_{\infty,\rho}) - \phi_i(\tilde{U}_{\infty,\rho})| \leq 4 \left(1 + \frac{k_{\max}^2}{\rho} \right) \frac{k_{\max}^2}{\rho^2} \left\| \tilde{k}(X_N, X_N) - k(X_N, X_N) \right\|_2 \frac{1 + \log n}{n}.$$

Proof. For notational simplicity, for any subset $S \subseteq N$ define

$$\alpha(S, k) := (k(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S, \quad \alpha(S, \tilde{k}) := (\tilde{k}(X_S, X_S) + \rho|S|I_{|S|})^{-1}Y_S.$$

Then, the difference between the two utilities is given by

$$\begin{aligned} |U_{\infty,\rho}(S) - \tilde{U}_{\infty,\rho}(S)| &= \frac{1}{|T|} \left| \|Y_T - k(X_T, X_S)\alpha(S, k)\|_2^2 - \|Y_T - k(X_T, X_S)\alpha(S, \tilde{k})\|_2^2 \right| \\ &\leq \frac{1}{|T|} \|2Y_T - k(X_T, X_S)(\alpha(S, k) + \alpha(S, \tilde{k}))\|_2 \|k(X_T, X_S)(\alpha(S, k) - \alpha(S, \tilde{k}))\|_2 \end{aligned}$$

We first bound the first factor. By the triangle inequality,

$$\|Y_T - k(X_T, X_S)\alpha(S, k)\|_2 \leq \|Y_T\|_2 + \|k(X_T, X_S)\|_2 \|\alpha(S, k)\|_2.$$

Since $k(X_S, X_S) \succeq 0$ and $\rho > 0$, we have $\|(k(X_S, X_S) + \rho|S|I_{|S|})^{-1}\|_2 \leq (\rho|S|)^{-1}$.

Moreover, by assumptions (i) and (ii), $\|k(X_T, X_S)\|_2 \leq k_{\max}\sqrt{|T||S|}$. Combining these bounds yields

$$\|Y_T - k(X_T, X_S)\alpha(S, k)\|_2 \leq \sqrt{|T|} + \frac{k_{\max}^2}{\rho} \sqrt{|T|}.$$

By the same argument, $\|Y_T - k(X_T, X_S)\alpha(S, \tilde{k})\|_2 \leq \sqrt{|T|} + \frac{k_{\max}^2}{\rho} \sqrt{|T|}$. Summing the two estimates gives

$$|U_{\infty, \rho}(S) - \tilde{U}_{\infty, \rho}(S)| \leq \frac{2}{\sqrt{|T|}} \left(1 + \frac{k_{\max}^2}{\rho}\right) \|k(X_T, X_S)(\alpha(S, k) - \alpha(S, \tilde{k}))\|_2.$$

Next, we bound $\|\alpha(S, k) - \alpha(S, \tilde{k})\|_2$. Let $A := k(X_S, X_S) + \rho m I_m$ and $B := \tilde{k}(X_S, X_S) + \rho m I_m$.

Using the resolvent identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$, we obtain

$$\begin{aligned} \|\alpha(S, k) - \alpha(S, \tilde{k})\|_2 &\leq \|A^{-1}\|_2 \|\tilde{k}(X_S, X_S) - k(X_S, X_S)\|_2 \|B^{-1}\|_2 \|Y_S\|_2 \\ &\leq \frac{\|\tilde{k}(X_S, X_S) - k(X_S, X_S)\|_2}{\rho^2 |S|^{3/2}}, \end{aligned}$$

where the last inequality follows from $\|A^{-1}\|_2, \|B^{-1}\|_2 \leq (\rho|S|)^{-1}$ and $\|Y_S\|_2 \leq \sqrt{|S|}$.

Consequently, we obtain the following bound for each subset $S \subseteq N$:

$$|U_{\infty, \rho}(S) - \tilde{U}_{\infty, \rho}(S)| \leq 2 \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{k_{\max}^2}{\rho^2 |S|} \|\tilde{k}(X_S, X_S) - k(X_S, X_S)\|_2. \quad (26)$$

$$\leq 2 \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{k_{\max}^2}{\rho^2 |S|} \|\tilde{k}(X_N, X_N) - k(X_N, X_N)\|_2. \quad (27)$$

where the second inequality follows from the monotonicity of the spectral norm under principal submatrices.

Applying Lemma E.1, and using the bound $\sum_{l=1}^n l^{-1} \leq 1 + \log n$, we obtain

$$|\phi_i(U_{\infty, \rho}) - \phi_i(\tilde{U}_{\infty, \rho})| \leq 4 \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{k_{\max}^2}{\rho^2} \|\tilde{k}(X_N, X_N) - k(X_N, X_N)\|_2 \frac{1 + \log n}{n}$$

□

Remark E.2. An important feature of Proposition 4.2 is that the bound does not depend on the smallest eigenvalues or the condition numbers of the principal submatrices $k(X_S, X_S)$. Instead, the approximation error is controlled only through the spectral error $\|\tilde{k}(X_N, X_N) - k(X_N, X_N)\|_2$.

Moreover, the bound is invariant under simultaneous rescaling of the kernel matrix and the regularization parameter. More precisely, if

$$k \mapsto ck, \quad \tilde{k} \mapsto c\tilde{k}, \quad \rho \mapsto c\rho$$

for some constant $c > 0$, then the right-hand side of the bound remains unchanged.

Remark E.3 (Best rank- r approximation). Suppose that the kernel approximation in Proposition 4.2 is chosen as the best rank- r approximation of $k(X_N, X_N)$. Since $k(X_N, X_N)$ is positive semidefinite, the approximation error in spectral norm is exactly the spectral tail:

$$\|\tilde{k}_{\text{best}}(X_N, X_N) - k(X_N, X_N)\|_2 = \lambda_{r+1}(k(X_N, X_N)).$$

Hence, Proposition 4.2 gives

$$|\phi_i(U_{\infty, \rho}) - \phi_i(\tilde{U}_{\infty, \rho}^{\text{best}})| \leq 4 \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{k_{\max}^2}{\rho^2} \frac{1 + \log n}{n} \lambda_{r+1}(k(X_N, X_N)).$$

This shows that the Shapley approximation error is controlled directly by the spectral tail of the kernel matrix.

Assume that the eigenvalues of the kernel matrix satisfy the power-law decay

$$\lambda_j(k(X_N, X_N)) \leq C j^{-\alpha}$$

for some constants $C > 0$ and $\alpha > 0$. Then the best rank- r approximation bound becomes

$$|\phi_i(U_{\infty, \rho}) - \phi_i(\tilde{U}_{\infty, \rho}^{\text{best}})| \leq 4C \left(1 + \frac{k_{\max}^2}{\rho}\right) \frac{k_{\max}^2}{\rho^2} \frac{1 + \log n}{n} (r+1)^{-\alpha}.$$

Using the efficiency axiom, individual Shapley values are naturally resolved at the $1/n$ -scale when $U(N) - U(\emptyset) = O(1)$. To make the above error smaller than this scale, it suffices to choose r such that

$$r \geq \left[4C \left(1 + \frac{k_{\max}^2}{\rho} \right) \frac{k_{\max}^2}{\rho^2} (1 + \log n) \right]^{1/\alpha} - 1.$$

For small ρ , the sufficient rank scales as $r \gtrsim \left(\frac{Ck_{\max}^4}{\rho^3} \log n \right)^{1/\alpha}$. Thus, faster eigenvalue decay, namely larger α , reduces the required rank, whereas smaller regularization ρ increases it.

Remark E.4 (Nyström approximation). One advantage of Proposition 4.2 is that it allows us to compare different low-rank kernel approximation methods through their spectral approximation errors.

Consider the Nyström approximation. Let $m < n$, and suppose that W is formed from m sampled columns of $k(X_N, X_N)$. After reordering the rows and columns, the kernel matrix can be written as

$$k(X_N, X_N) = \begin{bmatrix} W & K_{21}^\top \\ K_{21} & K_{22} \end{bmatrix}, \quad C = \begin{bmatrix} W \\ K_{21} \end{bmatrix}.$$

Let W_r denote the best rank- r approximation of W , where $r \leq m$. The truncated Nyström approximation is then given by

$$\tilde{k}_{\text{nys}}(X_N, X_N) := CW_r^\dagger C^\top.$$

According to (Drineas et al., 2005; Cortes et al., 2010), with probability at least $1 - \delta$,

$$\begin{aligned} \|\tilde{k}_{\text{nys}}(X_N, X_N) - k(X_N, X_N)\|_2 &\leq \|\tilde{k}_{\text{best}}(X_N, X_N) - k(X_N, X_N)\|_2 + \frac{n}{\sqrt{m}} k_{\max}^2 (2 + \log(1/\delta)) \\ &= \lambda_{r+1}(k(X_N, X_N)) + \frac{n}{\sqrt{m}} k_{\max}^2 (2 + \log(1/\delta)). \end{aligned}$$

Combining this bound with Proposition 4.2, we obtain, with probability at least $1 - \delta$,

$$|\phi_i(U_{\infty, \rho}) - \phi_i(\tilde{U}_{\infty, \rho}^{\text{nys}})| \leq 4 \left(1 + \frac{k_{\max}^2}{\rho} \right) \frac{k_{\max}^2}{\rho^2} \left[\lambda_{r+1}(k(X_N, X_N)) \frac{1 + \log n}{n} + \frac{1 + \log n}{\sqrt{m}} k_{\max}^2 (2 + \log(1/\delta)) \right].$$

The above bound shows why the Nyström approximation can be less suitable than the best rank- r approximation for individual Shapley value estimation. The best rank- r component contributes $\lambda_{r+1}(k(X_N, X_N)) \frac{1 + \log n}{n}$, whereas the Nyström approximation introduces an additional sampling-dependent term of order $\frac{1 + \log n}{\sqrt{m}}$, up to kernel- and confidence-dependent constants. Since this additional term is not on the natural $1/n$ -scale of individual Shapley values, it may dominate the bound unless m is sufficiently large. This can obscure the $1/n$ -scale resolution required for individual Shapley value estimation, even when the best rank error $\lambda_{r+1}(k(X_N, X_N))$ is already very small.

F. Computational complexity of FreeShap and LRFShap

Table 2. The computational complexity of FreeShap and Ours. Here S and T are the selected indices from the train and test dataset with $|S| = k$, and n is the number of the total train dataset. Also, we fix the eigen rank as r .

FreeShap vs. Ours						
	FreeShap			Ours		
	Operation	Complexity	Dominating	Operation	Complexity	Dominating
Preparation (1 Time)	-	-	-	Eigen Decom. Φ_{tr} Φ_{te}	$\mathcal{O}(n^3)$ $\mathcal{O}(nr^2)$ $\mathcal{O}(nr(T + r))$	$\mathcal{O}(n^2(n + T))$
w/o Caching	$K(X_S, X_S)^{-1}$	$\mathcal{O}(k^3)$		$K_{SS} = \Phi_S \Phi_S^\top$	$\mathcal{O}(k^2 r)$	
	$\beta = K(X_S, X_S)^{-1} Y_S$	$\mathcal{O}(k^2)$	$\mathcal{O}(Ck(k^2 + T))$	$L = \text{Cholesky}(K_{SS} + \lambda I)$	$\mathcal{O}(k^3)$	$\mathcal{O}(kr(k + T))$
	$K(X_T, X_S)^\top \beta$	$\mathcal{O}(T k)$	$(k < 500)$	$\alpha = \text{Chole.solve}(Y_S, L)$	$\mathcal{O}(k^2 C)$	$(k < r)$
	for i in range(C)	$C \times$		$\Phi_{te} \Phi_S \alpha$	$\mathcal{O}(T k(r + C))$	
w/ Caching	$K(X_S, X_S)^{-1}$ (Block Inverse)	$\mathcal{O}(k^2)$	$\mathcal{O}(Ck(k + T))$	$B = \Phi_S^\top \Phi_S$	$\mathcal{O}(kr^2)$	
	$\beta = K(X_S, X_S)^{-1} Y_S$	$\mathcal{O}(k^2)$	$(k \geq 500)$	$L = \text{Cholesky}(B + \lambda I)$	$\mathcal{O}(r^3)$	$\mathcal{O}(kr^2)$
	$K(X_T, X_S)^\top \beta$	$\mathcal{O}(T k)$		$R = \Phi_S^\top Y_S$	$\mathcal{O}(krC)$	$(k \geq r)$
	for i in range(C)	$C \times$		$W = \text{Chole.solve}(R, L)$	$\mathcal{O}(r^2 C)$	
				$\Phi_{te} W$	$\mathcal{O}(T rC)$	
				$B^{-1} = (\Phi_S^\top \Phi_S + \lambda I)^{-1}$ (Sherman-Morrison)	$\mathcal{O}(r^2)$	$\mathcal{O}(r^2 C)$
				$R = \Phi_S^\top Y_S$	$\mathcal{O}(krC)$	$k \geq r$
				$W = B^{-1} R$	$\mathcal{O}(Cr^2)$	
				$\Phi_{te} W$	$\mathcal{O}(Cr T)$	

Speedups in terms of computational complexity:

- Without caching: $\mathcal{O}(Ck(k^2 + |T|)) \rightarrow \mathcal{O}(kr(k + |T|))$ (when $k < r$) or $\mathcal{O}(kr^2)$ (when $k \geq r$). In fact, if $k < r$, our method is slower than FreeShap. However, this only occupies a small part of computation since r is small, e.g., $r = 0.2n$.
- With caching: $\mathcal{O}(Ck(k + |T|)) \rightarrow \mathcal{O}(Cr^2)$ ($k \geq r$).
- Eigen decomposition is required only once.